Figure 1: Fedora for Science

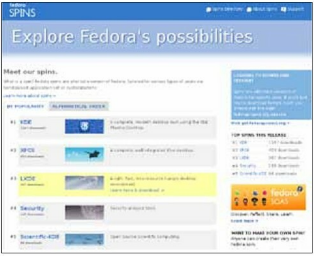

Figure 2: The spin page

Torque resource manager to enable you to set up a batch-processing system.

- **Editing, drawing and visualisation tools:** So you have simulated your grand experiments, and need to visualise the data, plot graphs, and create publication-quality articles and figures. The tools included to help you in this include *LaTex* compilers and the *Texmaker* and *Kile* editors, plotting and visualisation tools *Gnuplot, xfig, MayaVi, Dia* and *Ggobi* , and the vector graphics tool *Inkscape*.
- **Version control, back-up tools and document managers:** Version control and back-up tools are included to help you manage your data and documents better: *Subversion, Git* and *Mercurial* are available, along with the back-up tool *backintime*. Also included is a bibliography manager, *BibTool*.

Besides these four main categories, some of the other miscellaneous utilities include: *hevea*–the awesome LaTex-to-HTML converter, *GNU Screen* and *IPython*. The complete list of all the additional packages included in the spin can be found at *https://fedoraproject.org/wiki/Scientific_Packages_List*.

Fedora Spins

Now that we have taken a look at *Fedora Scientific*, let us explore the enabler behind it. What made Fedora Scientific possible is the *Fedora Spins* effort. Quoting from the website: "Fedora Spins are alternate versions of Fedora, tailored for various types of users, via hand-picked application sets or customisations" *(http://spins.fedoraproject.org/)*. As of the Fedora 16 release, there are nine custom spins—six of them highly customised for niche audiences like security professionals, designers, kids, researchers and robotics enthusiasts.

Creating a custom Fedora Spin is really easy. And unlike a lot of things in life, it is as easy to do it as it is to talk about it. A tool called *'livecd-creator' (http://fedoraproject.org/wiki/How_to_create_and_use_a_Live_CD)* is used to create a custom Fedora Spin. A configuration file called a *kickstart* file needs to be created first, where you specify the list of packages that you want to be installed. You can also specify various other custom configurations, such as the desktop icons, launchers, etc. You can take a look at the *kickstart* files for all the Fedora Spins at *http://bit.ly/sqZkVh*. If you have an idea for a custom spin, start by taking a look at one of these spins and then creating a *kickstart* file for yourself.

Once you have a *kickstart* file ready, you can use *livecd-creator* to create a customised Live ISO using the following code:

```
# livecd-creator --config=fedora-livecd-custom.ks \
--fslabel=Fedora-Live-Custom-CD -- cache=/var/cache/live
```

This will start the spinning process for your shiny new Fedora Spin. Once you have the ISO, you can write it to a USB stick using the '*dd*' command.

Where next?

First, if your interest in this article was to know more about Fedora Scientific, then head to *http://spins.fedoraproject.org/scientific-kde/* to download an ISO and try it for yourself. While you are at it, you may direct your queries and comments to the SciTech SIG mailing list at *https://admin.fedoraproject.org/mailman/listinfo/scitech*, or use the other forms of communication listed in the support tab.

However, if your interest was to know more about Fedora Spins in general, head over to the Fedora Spins page at *http://spins.fedoraproject.org/*, and learn more about the Fedora Spins process at *http://fedoraproject.org/wiki/SIGs/Spins*. Happy spinning! END

By: Amit Saha

The author is a part-time technical author with interests covering most of what is unknown. He blogs at *http://echorand.me/* and welcomes suggestions, comments and queries at *amitsaha.in@gmail.com*